

Extnded hint for 9.17. One needs to show that $(A'B') \nparallel (XP)$; otherwise, the first line in the proof does not make sense. By the transversal property, this means that

$$(*) \quad 2 \cdot \angle XPA' + 2 \cdot \angle PA'B' \neq 0.$$

This should be done later in the proof, right before we use point Y .

Instead of $2 \cdot \angle AXY \equiv 2 \cdot \angle AA'Y$, we should write

$$(**) \quad 2 \cdot \angle AXP \equiv 2 \cdot \angle PA'B'.$$

Since $\triangle XAP$ has a right angle at A , we have

$$\angle AXP + \angle XPA \equiv \pm \frac{\pi}{2}.$$

Since $2 \cdot \angle XPA' \equiv 2 \cdot \angle XPA$, $(**)$ implies that

$$2 \cdot \angle XPA' + 2 \cdot \angle PA'B' \equiv \pi,$$

and $(*)$ follows.

Additionally, we implicitly use the following identities:

$$2 \cdot \angle AXP \equiv 2 \cdot \angle AXY,$$

$$2 \cdot \angle ABP \equiv 2 \cdot \angle ABB',$$

$$2 \cdot \angle AA'B' \equiv 2 \cdot \angle AA'Y.$$